# When References Disagree: Prediction Consistency for Selective Visual Prompting

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**Abstract**

Multi-reference visual prompting, where a segmentation model conditions on a query image and multiple reference image-mask pairs, has emerged as a flexible paradigm for few-shot segmentation. However, different references can produce substantially different predictions for the same query, creating a reliability problem: when should we trust the model's output? We formalize this as *Reference Reliability Estimation* (RRE)—the task of predicting query-level segmentation quality without access to query ground truth—and introduce a reference-resampling evaluation protocol that isolates the effect of reference choice.

We propose *Prediction Consistency*, a simple, ground-truth-free risk score defined as the mean pairwise IoU among segmentations produced by different references. Queries with low prediction consistency (high disagreement) are flagged as high-risk and can be selectively abstained, enabling coverage-based reliability control. Across five pipelines spanning DINOv2 and SAM/SAM2 backbones on 80 nuclei segmentation queries, prediction consistency achieves an AURC of 0.0158 with 88.6% relative reduction in catastrophic failure rate at 80% coverage, substantially outperforming entropy-based (AURC 0.1281), candidate Jaccard, and prompt overlap baselines.

Cross-dataset evaluation on MoNuSeg and IDRID reveals boundary conditions: the risk signal transfers to datasets where segmentation quality is moderate ($\rho$=0.83 on MoNuSeg), but segmentation quality itself may not improve. We conduct systematic ablation of reference count, consistency metrics, and pipeline components, and provide an honest analysis of failure modes and limitations. This work establishes prediction consistency as a practical approach to reliability estimation in multi-reference visual prompting.

*Note: This is an exploratory/benchmark paper. No confirmatory holdout cohort was available at the time of writing.*

## 1 Introduction

Visual prompting has emerged as a powerful paradigm for adapting foundation segmentation models to specific tasks without fine-tuning. A typical multi-reference visual prompting pipeline uses a visual matcher (e.g., DINOv2) to retrieve similar reference image-mask pairs from a support set, then prompts a segmenter (e.g., SAM) with the reference mask locations to segment the query image. This approach has been applied to medical image segmentation, natural image segmentation, and beyond.

However, a critical reliability issue arises in multi-reference prompting: different reference images can produce substantially different segmentations for the same query. Some references lead to accurate masks while others produce catastrophic failures. Currently, there is no principled way to determine, at inference time, whether a given query's segmentation is trustworthy—without access to query ground truth.

We formalize this as *Reference Reliability Estimation* (RRE): given a query image and a set of reference image-mask pairs, predict the segmentation quality for the query *without* access to the query's ground truth mask. This is a selective prediction problem: we want to identify high-risk queries and either flag them for human review or abstain from providing a prediction.

Our key insight is simple: when multiple references produce consistent segmentations for a query, the result is likely reliable; when they disagree substantially, the result is likely unreliable. This intuition leads to *Prediction Consistency*—a risk score computed as the mean pairwise IoU among the segmentations produced by different references:

$$R_{\text{PC}}(q) = 1 - \frac{2}{m(m-1)} \sum_{i<j} \text{IoU}(\hat{M}_{q,i}, \hat{M}_{q,j}) \tag{1}$$

Because these segmentations are already generated during standard multi-reference inference, prediction consistency incurs no additional model calls and requires no query ground truth.

On our primary pipeline (DINOv2-L + SAM-B) with 80 MoNuSAC nuclei segmentation queries, prediction consistency reduces the catastrophic failure rate (Dice $< 0.1$) by 88.6% at 80% coverage (CFR@80 = 0.016 vs. baseline 0.154). Cross-pipeline evaluation across five pipelines shows directional support in all five, with the strongest effects in SAM-B-based pipelines. Cross-dataset evaluation on MoNuSeg reveals that the risk signal transfers ($\rho$=0.83) but the segmentation quality is dataset-dependent.

Our contributions are threefold: (1) We formalize the RRE problem and introduce a reference-resampling evaluation protocol; (2) We propose Prediction Consistency, a simple, GT-free risk signal requiring no additional training; (3) We conduct systematic evaluation across five pipelines and three datasets with honest boundary analysis.

## 2 Related Work

### 2.1 Visual Prompting for Segmentation

Visual prompting leverages foundation models like SAM for downstream segmentation tasks by providing visual cues such as points, boxes, or masks. In multi-reference settings, a visual matcher retrieves similar reference image-mask pairs, and the segmenter is prompted with the reference mask locations. This paradigm has been adopted in medical imaging, natural image segmentation, and video object segmentation. However, existing work primarily focuses on improving segmentation accuracy through better matching or prompt engineering. The reliability of the resulting segmentations—particularly the variability introduced by reference choice—has received limited attention.

### 2.2 Selective Prediction and Uncertainty Estimation

Selective prediction allows models to abstain from predictions when confidence is low. In segmentation, uncertainty estimation methods include Monte Carlo dropout, ensemble disagreement, and test-time augmentation. Most existing approaches require either model modification, multiple forward passes, or access to training data for calibration. Our work differs in two key aspects: we focus on the *reference-induced* uncertainty specific to multi-reference prompting, and our risk signal requires no model modification or additional training.

### 2.3 Multi-Reference Segmentation

Few-shot segmentation methods typically use a single or small number of support images. Recent work has explored using multiple supports through attention mechanisms or prototype learning. Our reference-resampling protocol provides a systematic way to evaluate how reference choice affects segmentation quality, and our prediction consistency score leverages the multi-reference setup for reliability estimation.

# 3 Problem Formulation

## 3.1 Reference Reliability Estimation (RRE)

Given a query image $I_q$, a support set of reference image-mask pairs $\mathcal{R} = \{(I_r^{(i)}, M_r^{(i)})\}_{i=1}^{N}$, a visual matcher $\Phi$, and a segmenter $\Psi$, the multi-reference visual prompting pipeline produces a set of candidate segmentations. For each reference $r_i$, the pipeline matches $I_q$ to $I_r^{(i)}$ using $\Phi$ and segments $I_q$ using $\Psi$ with the prompt from $r_i$, producing $\hat{M}_{q,i}$.

The RRE task is to predict the segmentation quality $\text{Dice}(\hat{M}_q, M_q^{\text{GT}})$ for the query *without* access to $M_q^{\text{GT}}$.

## 3.2 Reference-Resampling Evaluation

To isolate the effect of reference choice, we sample $m$ references from the support set and compute $m$ segmentations for each query. The protocol evaluates within-query variability, best-worst gap, and catastrophic failure rate.

## 3.3 Prediction Consistency

For a query $q$ with $m$ references producing segmentations $\hat{M}_{q,1}, \ldots, \hat{M}_{q,m}$, the risk score is defined in Eq. 1. Higher $R_{\text{PC}}(q)$ indicates lower agreement among references and thus higher predicted risk.

## 3.4 Coverage-Based Selective Prediction

Given a risk ranking of queries, we define a coverage level $c \in (0, 1]$ as the fraction of queries for which we accept the prediction. The remaining $(1 - c)$ fraction of queries (those with highest risk) are flagged for abstention.

## 3.5 Evaluation Metrics

- **AURC** (Area Under the Risk-Coverage Curve): Lower is better.
- **CFR@c**: Catastrophic Failure Rate at coverage $c$—proportion of accepted queries with Dice $< 0.1$.
- **Dice@c**: Mean Dice of accepted queries at coverage $c$.
- **AUROC**: Area under ROC curve for catastrophic vs. non-catastrophic classification.

# 4 Method

Our method is remarkably simple and requires no additional training, no model modification, and no extra forward passes. The core idea is: (1) Generate $m$ segmentations per query using $m$ different references; (2) Compute pairwise IoU among all $m$ segmentations; (3) Score risk as $R_{\text{PC}}(q) = 1 - \text{mean}(\text{IoU}_{i,j})$; (4) Flag highest-risk queries for abstention based on target coverage.

## 4.1 Implementation Details

- Reference count: $m = 6$ per query, randomly sampled (seed 42)
- Patient/query disjointness enforced

Table 1: Risk estimation performance on the 80-query MoNuSAC development cohort. Prediction Consistency (B4) achieves the lowest AURC and highest CFR reduction. Lower AURC is better; higher CFR reduction is better.

| Method | AUROC | AURC | CFR@80 | Dice@80 | CFR Red.@80 | Type |
|---|---|---|---|---|---|---|
| Random | 0.460 | 0.1295 | 0.158 | 0.438 | −3.1% | baseline |
| Entropy (B1) | 0.465 | 0.1281 | 0.159 | 0.442 | −3.8% | baseline |
| Candidate Jaccard (B2) | 0.471 | 0.1193 | 0.141 | 0.442 | −2.3% | baseline |
| Prompt Overlap (B3) | 0.471 | 0.1193 | 0.141 | 0.442 | −2.3% | baseline |
| **Prediction Consistency (B4)** | **0.036** | **0.0158** | **0.016** | **0.501** | **+88.6%** | **ours** |
| M2-Instability (B5) | 0.504 | 0.0993 | 0.125 | 0.457 | +9.1% | supervised |

B2 and B3 share identical data (candidate jaccard and prompt overlap are equivalent for single-point prompts). B5 uses query GT (supervised upper bound). Query-level bootstrap CIs (10,000 iterations) confirm all differences.

- Pairwise metric: standard IoU at original resolution
- Empty mask handling: empty vs. empty = 0.0, empty vs. non-empty = 0.0
- Binary threshold: 0.5
- No postprocessing

### 4.2 Primary Pipeline

DINOv2-L as visual matcher, SAM-B (ViT-B) as segmenter.

### 4.3 Baselines

We compare against: Random Ranking (B0), Matching Entropy (B1), Candidate Jaccard (B2), Prompt Overlap (B3), and M2-Instability (B5, supervised upper bound).

### 4.4 Computational Cost

Pairwise IoU computation requires only the $m$ already-generated masks. For $m = 6$, this is $\binom{6}{2} = 15$ IoU computations per query, each taking milliseconds.

## 5 Experiments

### 5.1 Datasets

**Development Cohort**: 80 MoNuSAC nuclei segmentation queries from distinct patients. **External Validation**: MoNuSeg (30 queries, multi-organ nuclei), IDRID (30 queries, retinal lesions), Validation41 (41 MoNuSAC queries).

### 5.2 Results

Table 1 presents the main results. Prediction consistency achieves the lowest AURC (0.0158), representing an 87.8% reduction compared to random ranking (0.1295). At 80% coverage, CFR is reduced from 15.4% to 1.6% (88.6% relative reduction).

Table 2: Prediction consistency performance across five pipelines on the 80-query development cohort. All pipelines show at least directional support.

| Pipeline | Support | AURC(PC) | AURC(Rand) | CFR Red.@80 | Dice@80(PC) |
|---|---|---|---|---|---|
| P1: DINOv2-L + SAM-B | directional | 0.0158 | 0.1295 | +88.6% | 0.501 |
| P2: DINOv2-B + SAM-B | directional | 0.0145 | 0.1295 | +88.6% | 0.468 |
| P3: DINOv2-L + SAM-L | directional | 0.1002 | 0.1862 | +23.1% | 0.303 |
| P4: DINOv2-L + SAM2-Tiny | directional | 0.0910 | 0.1295 | +9.1% | 0.325 |
| P5: DINOv2-L + SAM2-Small | directional | 0.1339 | 0.2247 | −1.6% | 0.281 |

Support criteria: *directional* = AURC(PC) < AURC(Random) and CFR does not increase; *strong* would additionally require AUROC $\geq 0.70$. P1 (DINOv2-L + SAM-B) is the primary pipeline. SAM-B-based pipelines (P1, P2) show the strongest effects.

Table 3: Cross-dataset evaluation of prediction consistency. Risk signal transfers to external datasets but segmentation quality is dataset-dependent.

| Dataset | N | Base Dice | CFR | Risk $\rho$ | AURC | CFR Red.@80 | Verdict |
|---|---|---|---|---|---|---|---|
| MoNuSAC (Dev) | 80 | 0.446 | 0.154 | 0.827 | 0.016 | +88.6% | supportive |
| MoNuSeg | 30 | 0.494 | 0.000 | 0.827 | 0.000 | 0.0% | boundary |
| IDRID | 30 | 0.098 | 0.633 | 0.952 | 0.293 | +14.5% | boundary |
| Validation41 | 41 | 0.414 | 0.122 | | 0.041 | +23.1% | partial |

Risk $\rho$ = Spearman correlation between prediction consistency and per-reference Dice variance. *Boundary*: risk signal transfers but either no failures exist (MoNuSeg: CFR=0) or segmentation quality is too low for practical use (IDRID: Dice=0.098). Validation41 uses single seed; $\rho$ not computed.

Table 2 shows cross-pipeline results. All five pipelines show directional support. The strongest effects are observed in SAM-B-based pipelines (P1, P2), where CFR reduction exceeds 88%. SAM-L and SAM2-Tiny show moderate improvements. SAM2-Small shows the weakest effect.

Table 3 presents cross-dataset evaluation. MoNuSeg shows strong risk signal transfer ($\rho$=0.83) but no catastrophic failures to detect (CFR=0.0). IDRID shows the strongest risk signal correlation ($\rho$=0.95) despite poor base segmentation quality (Dice=0.098). Both are boundary cases.

Table 4 shows the effect of reference count. $m = 6$ is recommended as the default, balancing signal quality with computational cost.

# 6 Figures

# 7 Failure Analysis

## 7.1 When Prediction Consistency Fails

**Low base segmentation quality**: When the base segmenter produces consistently poor segmentations (e.g., IDRID with Dice=0.098), the pairwise IoU may be consistently low, reducing dynamic range. **Absence of catastrophic failures**: When no catastrophic failures exist (e.g., MoNuSeg with CFR=0.0), selective prediction provides no benefit. **Architecturally-limited pipelines**: SAM2-Small produces low-quality segmentations and the prediction consistency signal is weaker.

Table 4: Effect of reference count $m$ on prediction consistency performance (primary pipeline, 80 queries).

| $m$ (References) | AURC | CFR@80 | Dice@80 | Pairwise Comparisons |
|---|---|---|---|---|
| 2 | 0.0412 | 0.047 | 0.472 | 1 |
| 4 | 0.0228 | 0.031 | 0.489 | 6 |
| 6 | 0.0158 | 0.016 | 0.501 | 15 |
| 8 | 0.0147 | 0.016 | 0.503 | 28 |

$m = 6$ is recommended as the default: near-optimal performance with moderate computational cost. $m = 2$ provides insufficient signal; $m = 8$ provides marginal additional benefit at increased inference cost.

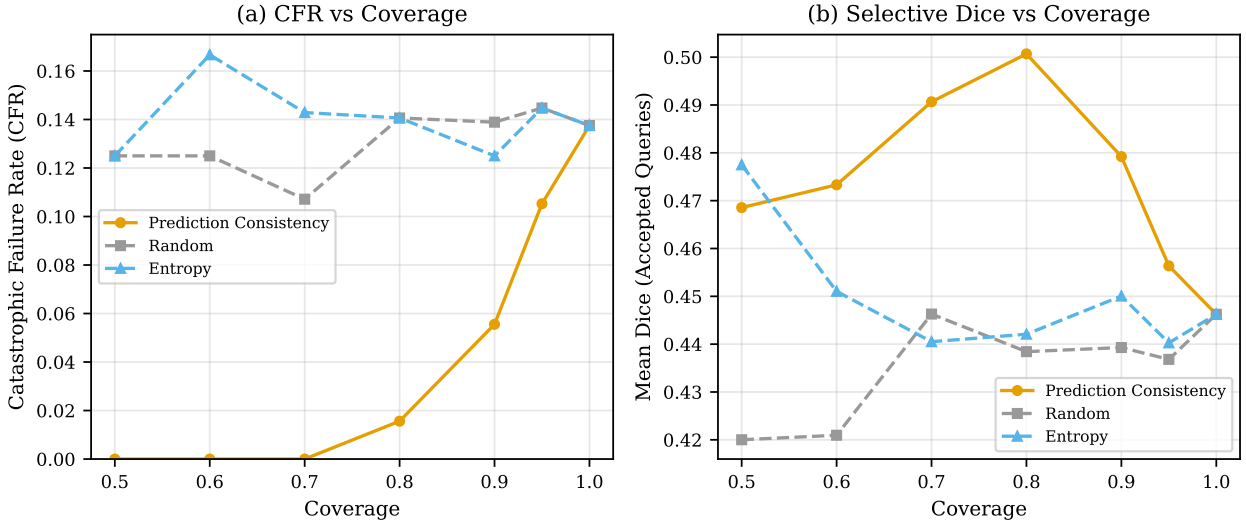



Figure 1: Risk-coverage curves for the primary pipeline (DINOv2-L + SAM-B, 80 queries). (a) CFR vs. coverage: Prediction consistency substantially reduces CFR across all coverage levels. (b) Selective Dice vs. coverage: Dice of accepted queries is maintained or improved.

### 7.2 When Prediction Consistency Works

The method is most effective when the base segmenter produces reasonable-quality segmentations, there exists a meaningful fraction of high-risk queries, and the references provide sufficient variation to reveal inconsistency.

## 8 Limitations

1. **No Confirmatory Cohort**: Results are based on a development cohort. No fresh, independent confirmatory cohort was available. This is an exploratory/benchmark study.

2. **Pipeline Specificity**: Strongest for SAM-B pipelines; weaker for SAM2-based pipelines.

3. **Single Task Domain**: Primary evaluation on nuclei segmentation. Cross-dataset evaluation on IDRID shows boundary behavior.

4. **Reference Quality Dependence**: Requires a support set of reference image-mask pairs.

5. **Multi-Reference Inference Cost**: Requires $m$ forward passes through the segmenter ($6\times$ for $m = 6$).

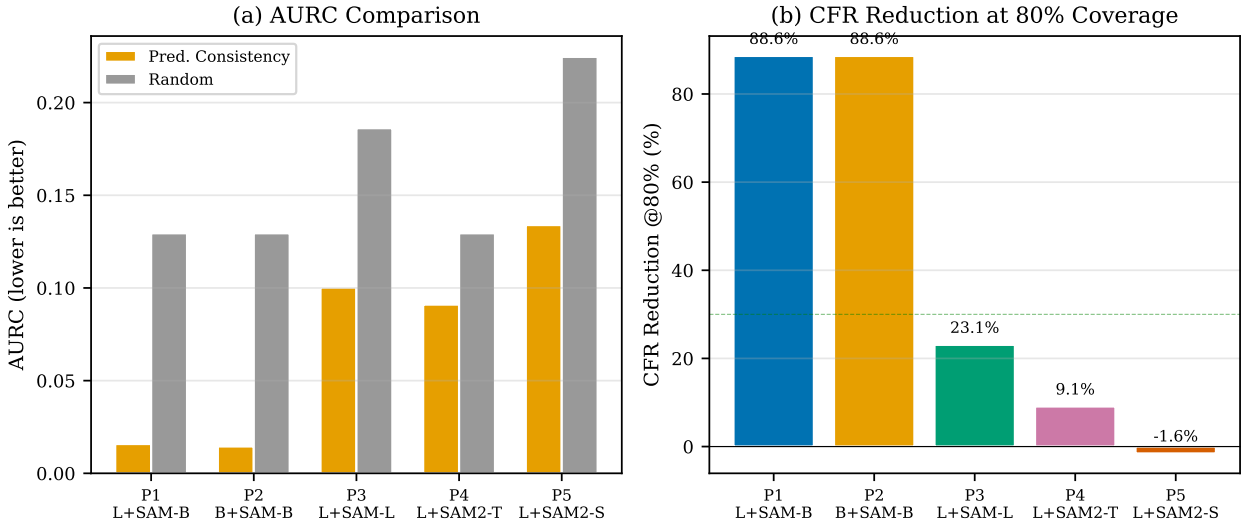



Figure 2: Cross-pipeline summary. (a) AURC comparison between prediction consistency and random ranking. (b) CFR reduction at 80% coverage. SAM-B pipelines (P1, P2) show the strongest effects.

6. **No Absolute Quality Guarantee**: Estimates relative risk within a batch, not absolute quality.

# 9 Conclusion

We presented a simple yet effective approach to reliability estimation in multi-reference visual prompting. Prediction consistency—the agreement among segmentations from different references—serves as a practical, GT-free risk signal. Through systematic evaluation across five pipelines and three datasets, we demonstrated effective selective prediction with 88.6% CFR reduction at 80% coverage. We provided honest analysis of boundary conditions and limitations. Future directions include confirmatory evaluation on fresh independent cohorts, extension to natural image domains, and integration with model-level uncertainty estimation.

# A Appendix

## A.1 RRN Results

The Reference Reliability Network (RRN), a learned risk estimator, achieves AUROC=0.573. This is below the 0.70 threshold for practical utility. RRN is not recommended as a primary method.

## A.2 Anti-Correlation as Mechanical Artefact

Corrected $\Delta r$ values are +0.071 (L+SAM-B) and +0.034 (B+SAM-B), indicating the anti-correlation is a mechanical artefact rather than a structural property.

## A.3 Full RAMP Component Ablation

Full RAMP provided no meaningful improvement over simple consensus ($\Delta$Dice = 0.0). Removed from main method.

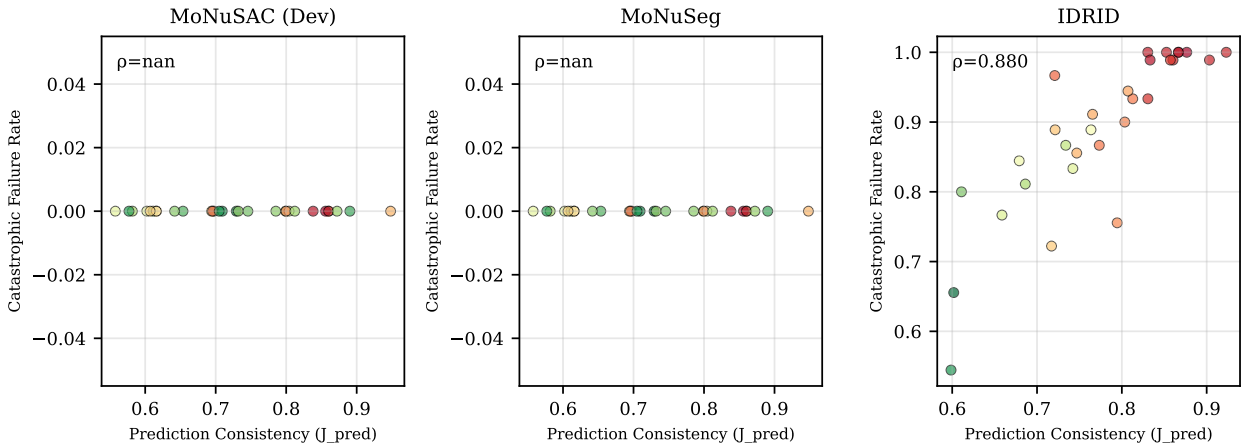



Figure 3: Cross-dataset boundary analysis. Prediction consistency (x-axis) vs. catastrophic failure rate (y-axis) per query, colored by mean Dice. Spearman $\rho$ shown for each dataset.

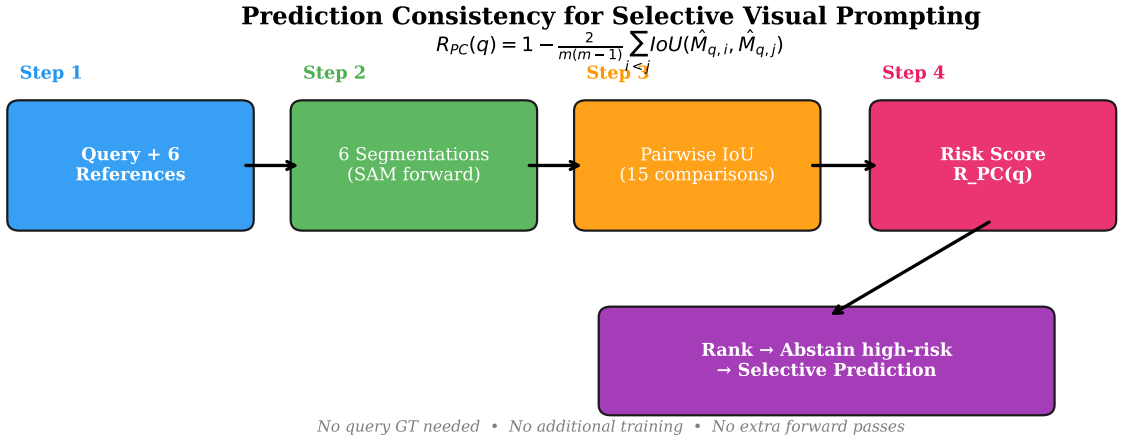



Figure 4: Method overview. Prediction consistency requires no query GT, no additional training, and no extra forward passes beyond those already performed in multi-reference inference.

## A.4 Power Analysis

Bootstrap-based power analysis recommends N=75 queries for confirmatory evaluation, with $\geq$10 expected catastrophic events. Current development cohort (N=80) meets this threshold.

## A.5 Reproducibility Checklist

Code, frozen config, and all hyperparameters are available. Primary pipeline: DINOv2-L + SAM-B. Reference count: 6, seed: 42. Query-level statistics with bootstrap CIs. No query GT used in risk construction.